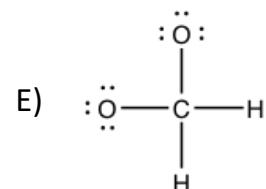
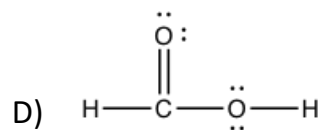
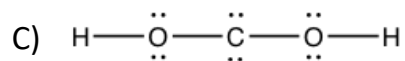
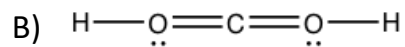
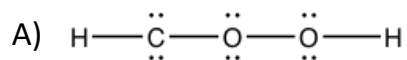


1. Identify the correct Lewis structure for formic acid, HCOOH (or CH<sub>2</sub>O<sub>2</sub>).



2. Which molecule is most likely to have *equivalent* resonance structures?

- A) SO<sub>2</sub>
- B) N<sub>2</sub>
- C) CO<sub>2</sub>
- D) F<sub>2</sub>
- E) CO

3. In which molecule would you expect deviations from the ideal bond angles?

- I. XeF<sub>4</sub>
- II. XeO<sub>2</sub>F<sub>2</sub>
- III. ICl<sub>2</sub><sup>+</sup>

- A) II only
- B) III only
- C) Both I and II
- D) Both II and III
- E) All of I, II, and III

4. Which species are tetrahedral?

- I.  $\text{SiCl}_4$
- II.  $\text{SeF}_4$
- III.  $\text{XeF}_4$
- IV.  $\text{Cl}_4$

- A) Both I and II
- B) Both II and III
- C) Both I and IV
- D) Both III and IV
- E) All of I, II, III, and IV

5. Which statement is true regarding internuclear distance, potential energy, and bond formation?

- A) Attraction between atoms increases potential energy.
- B) The amount of energy released when a bond is formed is determined by the internuclear distance that balances attraction and repulsion between atoms.
- C) Increasing internuclear distance always increases potential energy.
- D) Bond length is directly associated with the distance between atoms at which attraction first occurs.
- E) Bond strength is directly associated with the distance at which there is no repulsion between atoms.

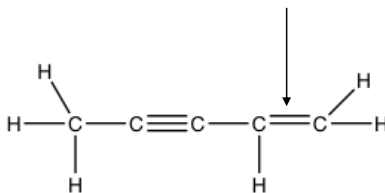
6. Which molecule must be nonpolar but contains polar covalent bonds?

- A)  $\text{XeCl}_4$
- B)  $\text{XeO}_2\text{Cl}_2$
- C)  $\text{PCl}_3\text{F}_2$
- D)  $\text{SF}_4$
- E)  $\text{CCl}_2\text{F}_2$

7. Which combination of orbitals *cannot* be used to make a sigma bond?

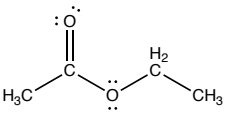
- A)  $sp^3$  hybrid orbital and  $sp^3$  hybrid orbital
- B)  $s$  orbital and  $s$  orbital
- C) two  $p$  orbitals oriented in the internuclear axis
- D) an  $s$  orbital and a  $p$  orbital oriented in the internuclear axis
- E) All of these orbitals can be used to make sigma bonds.

8. Which answer option best describes the double bond indicated by the arrow in the molecule image?



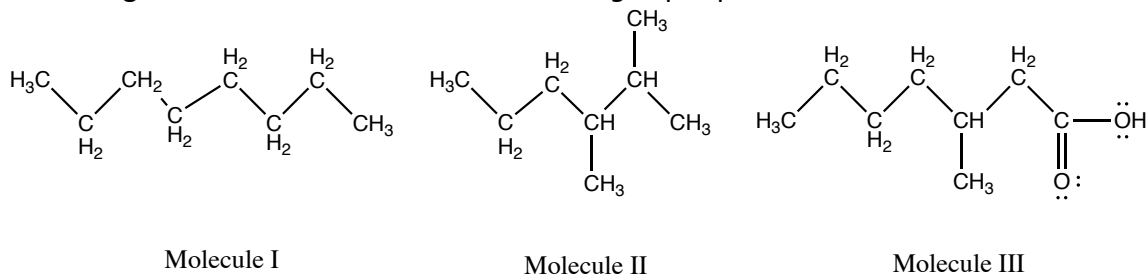
- A) Two sigma bonds formed between the  $sp^2$  hybrid orbitals of each C atom.  
B) Two pi bonds formed between unhybridized  $2p$  orbitals of each C atom.  
C) A sigma bond formed between the  $sp^2$  hybrid orbitals of each C atom and a pi bond formed between unhybridized  $2p$  orbitals of each C atom.  
D) A sigma bond formed between the  $2p$  orbitals of each C atom and a pi bond formed between hybridized  $sp^2$  orbitals of each C atom.  
E) Two pi bonds, one formed between the  $sp^2$  hybrid orbitals of each C atom and one formed between unhybridized  $2p$  orbitals of each C atom.
9. Which statement is true regarding bond lengths and strengths?
- A) More energy is released when a triple bond forms between two atoms than when a double bond forms between the same two atoms.  
B) A triple bond between two atoms is longer than a double bond between the same two atoms.  
C) More energy is released when a single bond between two atoms is broken than when a double bond between the same two atoms is broken.  
D) A single bond is longer and stronger than a double bond formed between the same two atoms.  
E) None of these statements are true.
10. Identify the *primary* intermolecular force between molecules of CH<sub>2</sub>F<sub>2</sub>.
- A) Ion-ion interactions  
B) Ion-dipole interactions  
C) Hydrogen bonding interactions  
D) Dipole-dipole interactions  
E) London dispersion forces

11. The dissolution of sodium nitrate in which solvent involves ion-dipole interactions?

- I.  $\text{H}_2\text{O}$   
II.  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$   
III. 

- A) Both I and II  
B) Both II and III  
C) All of I, II, and III  
D) I only  
E) Both I and III

12. Arrange the molecules in order of *increasing* vapor pressure.



- A)  $\text{II} < \text{I} < \text{III}$   
B)  $\text{III} < \text{II} < \text{I}$   
C)  $\text{I} < \text{III} < \text{II}$   
D)  $\text{II} < \text{III} < \text{I}$   
E)  $\text{III} < \text{I} < \text{II}$

13. What is the total change in enthalpy when 50.0 g of an unknown substance is heated from the solid at 90.7 to gas at 111.7 K? The entire process occurs at constant pressure.

Boiling point = 111.7 K

Freezing point = 90.7 K

$\Delta H_{\text{fus}} = 58.6 \text{ J/g}$

$\Delta H_{\text{vap}} = 511 \text{ J/g}$

$C_{\text{s (gas)}} = 2.09 \text{ J/g}\cdot^{\circ}\text{C}$

$C_{\text{s (liquid)}} = 1.12 \text{ J/g}\cdot^{\circ}\text{C}$

$C_{\text{s (solid)}} = 1.03 \text{ J/g}\cdot^{\circ}\text{C}$

- A)  $1.18 \times 10^4 \text{ J}$   
B)  $2.56 \times 10^4 \text{ J}$   
C)  $2.93 \times 10^4 \text{ J}$   
D)  $3.00 \times 10^4 \text{ J}$   
E)  $4.11 \times 10^5 \text{ J}$
14. Which statement is true when silicon is doped with gallium?
- A) Electrons are added to create a donor band that is closer in energy to the conduction band, thus increasing conductivity.  
B) Holes are added to create an acceptor band that is farther in energy from the conduction band, thus decreasing conductivity.  
C) Electrons are added to create a donor band that is farther in energy from the conduction band, thus decreasing conductivity.  
D) Holes are added to create an acceptor band that is closer in energy to the valence band, thus increasing conductivity.  
E) There is no change in conductivity.
15. What is the average rate of production of C in the following reaction when A is consumed at an average rate of 0.200 M/s?
- $$2 \text{ A} + \text{B} \rightleftharpoons 3 \text{ C}$$
- A) 0.100 M/s  
B) 0.133 M/s  
C) 0.200 M/s  
D) 0.300 M/s  
E) 0.333 M/s

16. Determine the rate law for the reaction  $3 A + B \rightarrow 2 C$  given the data provided.

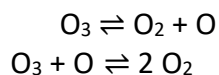
| Experiment | $[A]_{\text{initial}} (M)$ | $[B]_{\text{initial}} (M)$ | Rate (M/s) |
|------------|----------------------------|----------------------------|------------|
| 1          | 2.40                       | 1.72                       | 0.68       |
| 2          | 2.40                       | 3.44                       | 5.44       |
| 3          | 4.80                       | 5.16                       | 36.72      |

- A) Rate =  $k[A][B]^3$
- B) Rate =  $k[A]^3[B]^2$
- C) Rate =  $k[A]^3[B]$
- D) Rate =  $k[A][B]^{2.5}$
- E) Rate =  $k[A]^2[B]^2$

17. The reaction  $2 A \rightarrow B$  has  $k = 0.461 \text{ M}^{-1}\text{s}^{-1}$ . What is the concentration after 1.5 minutes if the initial concentration of a sample is 1.0 M?

- A) 0.024 M
- B) 0.096 M
- C) 0.32 M
- D) 0.40 M
- E) 0.59 M

18. Ozone decomposition is thought to occur by a two-step process:



Identify any reaction intermediate(s) and the overall reaction.

- A)  $\text{O}_3$  is the intermediate;  $2 \text{O}_3 \rightleftharpoons 3 \text{O}_2$
- B)  $\text{O}$  is the intermediate;  $2 \text{O}_3 \rightleftharpoons 3 \text{O}_2$
- C)  $\text{O}_2$  is the intermediate;  $2 \text{O}_3 + \text{O} \rightleftharpoons 3 \text{O}_2 + \text{O}$
- D)  $\text{O}$  is the intermediate;  $2 \text{O}_3 + \text{O} \rightleftharpoons 3 \text{O}_2 + \text{O}$
- E)  $\text{O}_2$  is the intermediate;  $2 \text{O}_3 \rightleftharpoons 3 \text{O}_2$

19. According to collision theory, under which circumstances may a collision between molecules *not* result in a chemical reaction?

- A) When collisions occur with insufficient kinetic energy.
- B) When molecules are not oriented properly.
- C) When molecules are in the gas phase.
- D) Both A and B.
- E) None of A, B, or C. All collisions between molecules result in chemical reactions.

20. Data collected in a laboratory experiment was used to create a graph of  $\ln k$  versus  $1/T$  ( $T$  in Kelvin). The slope of the resulting line is  $m$ . Which answer option represents the activation energy for the reaction used to collect the data?

- A)  $E_a/R$
- B)  $-E_a/R$
- C)  $mR$
- D)  $-mR$
- E)  $\ln A$

21. Which form of the exam do you have?

- A)
- B)